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REGULAR ARTICLES

Risk factors for clinical ketosis and association with milk production and reproduction variables in dairy cows in a hot environment

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Abstract

The aims of the present study were to investigate (1) the risk factors that influence the occurrence of clinical ketosis (CK; blood β-hydroxybutyrate > 3.0 mmol/L) and (2) to determine the influence of subclinical ketosis (SCK; 1.2 ≤ β-hydroxybutyrate ≤ 2.9 mmol/L) and CK on reproductive performance and milk yield in high-yielding Holstein cows in a hot environment. Cows ($n = 345$) were blood sampled from 6 to 15 days postpartum for β-hydroxybutyrate (BHB) determination with a hand-held meter. Cows calving in spring had 3.7 increased odds of having CK (20.7% incidence) than cows calving in summer (3.9% incidence) and autumn (9.4% incidence). Temperature-humidity index < 83 was associated with 1.6 times higher risk for CK compared with cows calving in warmer days. First-service conception rate was 12 and 16 percentage point higher ($P < 0.05$) in nonketotic cows compared with cows with SCK and CK, respectively. Actual 305-day milk yields for healthy, SCK, and CK cows were 9991 ± 1411, 10,123 ± 1442, and 10,386 ± 1435 kg (mean ± SD), respectively, with no difference ($P > 0.05$) between groups. In conclusion, this study documented that ketosis was seasonal with lower incidence of this metabolic disease during hot seasons and with increased ambient temperature at calving. Also, 305-day milk yield of Holstein cows was not related to blood BHB content early in lactation in this hot environment. However, elevated circulating BHB was negatively associated with conception rate at first service and fetal losses.

Keywords Subclinical ketosis · β-hydroxybutyrate · Milk yield · Conception rate · Services per pregnancy

Introduction

In early lactation, dairy cows are unable to meet their energy demand by the mammary gland from feed intake alone (VandeHaar and St-Pierre 2006), because dry matter intake (DMI) decreases by around 30% in the last 3 weeks of gestation (Hayirli et al. 2002). This limits the availability of energy during a time of increased demand of this nutrient for milk synthesis and triggers the mobilization of fat from the adipose tissue to use it as an energy fuel (Adewuyi et al. 2005). Excessive ketogenesis and low DMI may result in hyperketonemia in dairy cows.

Several studies have shown that increased blood concentrations of β-hydroxybutyrate (BHB) are associated with reduced conception rate (Tillard et al. 2007), pregnancy at first service (Walsh et al. 2007), and milk yield in early lactation (Duffield et al. 2009). However, the associations between increased levels of BHB and reproductive disorders have been inconsistent and the overall association between these variables is still inconclusive (McArt et al. 2013a).

The majority of the studies to determine the relationship between blood BHB at the beginning of lactation and milk yield and reproductive disorders have been carried out in temperate zones. In hot environments, these associations remain almost completely unstudied. Therefore, the objectives of this study were (1) to estimate the within-herd incidence of subclinical ketosis (SCK) and clinical ketosis (CK) for a Holstein herd in a hot environment, (2) to evaluate the association between SCK and CK on milk production and reproductive performance of cows, and (3) to determine the risk factors for the occurrence of CK (blood BHB > 3.0 mmol/L) at 6 to 15 days in milk (DIM).

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Material and methods

Study herd, housing, and feeding

Three hundred and forty-five Holstein cows were selected from an intensive 2000 Holstein cow farm in northern Mexico (26° N, mean annual temperature 24.6 °C). Cows were 1 to 7 years old with a body condition score (BCS) between 2.25 and 4.0 at calving (5-point scale; Roche et al. 2004). Cows were fed a TMR formulated to meet or exceed the predicted requirements of nutrients of a lactating Holstein cow yielding 40 kg of milk/day and 640 kg of body weight (BW; NRC 2001). The TMR consisted of 50% forage and 50% concentrate for ad libitum intake. Cows were housed in large, open-air, dirt-floor pens and they had unlimited access to fresh water. BCS of cows was recorded by the same person at calving.

Reproductive management

All cows were routinely vaccinated against diseases that affect the reproductive functions. The voluntary waiting period was set at 50 DIM. Detection of spontaneous estrus signs was made visually twice a day and cows were inseminated according to the AM–PM rule. All cows were examined for pregnancy status by transrectal palpation per rectum at 45 ± 3 days post-artificial insemination (AI). Cows not conceiving to the first service were reexamined for pregnancy after each subsequent insemination until 450 DIM.

Blood collection and assay of BHB

Blood samples were collected once from fresh cows between 6 to 15 DIM (SD = 2.1), from the coccygeal vein using a 0.5-mL tuberculin syringe and 27-gauge, 1.27-cm needle within 3 h of the morning feeding, immediately after the cows were milked. BHB was determined with a hand-held meter (Precision Xtra BHBA test, Abbott Laboratories) at cow side. A blood BHB level of 1.2 to 2.9 mmol/L was used as a cutoff for determining SCK. Blood values ≥ 3.0 mmol/L indicated CK (McArt et al. 2013b).

Milk yield and composition

Cows were milked two times daily (0400 and 1600 h). Milk yields were recorded every 2 weeks at each milking. Milk from both morning and afternoon milking were added. The morning and afternoon milk samples collected around 15 days postpartum were pooled to one composite sample for milk fat, protein, and lactose determination using near-infrared spectroscopy (Bentley 2000 Infrared Milk Analyzer, Bentley Instruments, Chaska, MN).

Meteorological recordings

For the duration of the study, daily maximum temperatures and relative humidity were obtained from a meteorological station located 1.5 km from the dairy barn. Climatological data from the farm were not available to confirm applicability from the meteorological station. This information was used to calculate the following temperature-humidity index: THI = (0.8 × maximum temperature) + [(% relative humidity/100) × (maximum temperature − 14.4)] + 46.4.

Statistical analyses

Cows that were eliminated ($n = 62$) during the post-calving period with some measurements recorded until the day they left the herd were included in the analysis. The final data set for reproductive and milk yield records included 283 cows (100 primiparous and 183 multiparous). Cows were characterized as first or second lactation and greater. BCS at calving was categorized into two groups: < 3.3 and greater than 3.3. Classes for THI at calving were defined as less or greater than 83 units. The records for milk fat were grouped into two classes: less or greater than 3.5%. Milk protein was grouped into two categories: lesser or greater than 3.0%. The milk lactose content was classified as being ≤ 4.6 or > 4.6%. The protein to fat ratio was categorized as lower or greater than 1.2. Season of calving was classed as spring (March 18 to June 21), summer (June 22 to September 21), and autumn (September 22 to December 29).

Descriptive statistics were obtained with the FREQ procedure of SAS (SAS Inst. Inc., Cary, NC). Backward elimination logistic regression was performed using $P = 0.10$ for leaving and reentering variables to create the final model. Using the final resultant model, all biologically plausible two-way interactions were examined. Only those variables retained in the final multivariable models are presented. The model included the following potentially explanatory variables: BCS, THI at calving, parity, milk fat, protein, and lactose content at blood collection, protein to fat ratio, and season of calving.

Pregnancy, fetal losses, and culling data were analyzed using the GENMOD procedure of SAS. The model statement for these dependent variables contained the effect of blood BHB levels; parity and BCS were included as covariates.

Data on milk yield and composition as well as various reproductive intervals were analyzed according to a completely randomized design using PROC MIXED procedure of SAS. If main effects were significant, means were compared using the PDIFF option in SAS. The effect of the occurrence of SCK or CK on the number of services per conception was evaluated by the npar1way procedure of SAS. For all analyses, significance was set at $P \leq 0.05$.

Results

The median blood BHB for all cows included in this study was 0.7 mmol/L, with a range of 0.3 to 6.3 mmol/L. The incidence of SCK in early lactation was 19.1% (66/345) whereas 10.1% (35/345) of cows experienced CK following calving. Cows whose parturition occurred in spring had trice the risk of clinical ketosis as had cows calving in summer or autumn (Table 1). Cows calving with a THI < 83 were 1.6 times more likely to suffer CK than cows calving in days with milder ambient temperatures.

Milk yield

Cows suffering SCK and CK did not differ in 305-day milk production when compared with nonketotic cows (Table 2). Peak yield of cows with SCK and CK was nearly 4 kg higher ($P < 0.05$) compared with nonketotic cows. The milk fat, protein, and lactose contents measured at 6–15days postpartum did not differ between hyperketonemic and healthy cows.

Reproductive performance

More cows became pregnant ($P < 0.01$) at the first AI for the healthy group compared with ketotic cows (Table 3); however, overall pregnancy rates (all services) did not differ between groups. Days to first AI were shorter ($P < 0.05$) for the nonketotic cows than for cows experiencing SCK or CK. Higher fetal losses ($P < 0.05$) were observed in cows with SCK at 6 to 15 DIM than nonketotic or cows with CK.

Discussion

This study showed that the incidence of SCK in postpartum Holstein cows in a hot environment was lower than the 6.9–47% incidence in European dairy farms (Vanholder et al. 2015), cumulative lactation incidence of 26.4–55.7% in US farms (McArt et al. 2012; Gordon et al. 2017), and 27.7% in Korea (Shin et al. 2015), when using blood samples for the measurement of blood BHB concentrations. One possible reason for the lower incidence of hyperketonemia in the present study compared with dairy farms in cooler climates is that heat-stressed cows use an homeorhetic mechanism leading to reduced fat mobilization from adipose tissue coupled with a reduced responsiveness to lipolytic stimuli (Baumgard and Rhoads 2013), which seems to reduce the occurrence of hyperketonemia. It is noteworthy to mention that comparing the incidence of ketosis is problematic because of the ample variety of conditions are existing when identifying this condition (different DIM for SCK diagnosis, parity of cows, cutpoints for defining ketosis, milk yield potential, blood or milk evaluation for ketone bodies, implementation of SCK prevention strategies, etc.).

This analysis indicated that cows with parturition during the hottest seasons (summer and autumn) had a lower risk of CK. Also, the risk of developing hyperketonemia was lower with higher THI at calving. It was expected an increased incidence of SCK and CK during the warmer seasons and in cows experiencing extreme heat stress at parturition. This is because in addition to environmental heat stress, the transition from the dry period to lactation in dairy cows implies a massive lipolysis as the cow is unable to consume sufficient nutrients to meet the energy requirements for milk synthesis. This leads to a negative energy balance (Bell and Bauman 1997), and the increased fat catabolism from adipose stores can result in ketosis (Drackley et al. 2005).

Similar to what it was found in the present study, odds of hyperketonemia remained low from July through November in dairy cattle in Ontario (Tatone et al. 2017). Likewise, lower ketosis risks were determined in heat-stressed than non-stressed dairy cows during early lactation in Croatia (Gantner et al. 2016). The reason for seasonality in SCK and CK incidence is unclear. Also, the lower occurrence of hyperketonemia in heat-stressed cows as compared with non-stressed cows is intriguing. A possible explanation for this response is that heat stress acclimation is accomplished by alteration in homeostatic responses (Horowitz 2002) which includes teleophoretic/homeorhetic processes including an altered endocrine status that affects target tissue responsiveness to environmental stimuli. Additionally, heat-stressed cows display a reduced hepatic growth hormone (GH) receptors and this

Table 1 Evaluation of the risk of clinical ketosis (blood β-hydroxybutyrate > 3.0 mmol/L)

Variable	Incidence (%)	Odds ratio (OR)	95% CI OR
Season			
Spring vs. summer and autumn	20.7 (18/87)	3.7	1.8–7.6
Summer vs. spring and autumn	3.9 (5/130)	0.3	0.1–0.7
Autumn vs. spring and summer	9.4 (12/128)	0.9	0.4–1.8
Temperature-humidity index			
< 83	14.1 (27/192)	1.6	1.0–2.5
> 83	5.2 (8/153)	Reference	

Table 2 Milk production variables for Holstein cows diagnosed with subclinical or clinical ketosis in a large commercial dairy operation in a hot arid environment

Milk variables	No ketosis ($n = 201$)	Subclinical ketosis ($n = 53$)	Clinical ketosis ($n = 28$)
Days to peak	88 ± 45	92 ± 46	92 ± 52
Peak milk yield, kg	40 ± 7^{a}	44 ± 8^{b}	44 ± 6^{b}
305-day milk yield, kg × 1000	9.99 ± 1.41	10.12 ± 1.44	10.39 ± 1.44
Milk fat, %*	3.6 ± 0.6	3.6 ± 0.6	3.6 ± 0.5
Milk fat yield, kg/lactation	360 ± 60	364 ± 61	374 ± 53
Milk protein, %*	3.0 ± 0.3	3.0 ± 0.2	2.9 ± 0.2
Milk protein yield, kg/lactation	300 ± 30	304 ± 21	301 ± 21
Milk lactose, %*	4.7 ± 0.3	4.6 ± 0.3	4.6 ± 0.3
Milk lactose yield, kg/lactation	470 ± 29	466 ± 30	478 ± 31
Milk fat to protein ratio	1.2 ± 0.2	1.2 ± 0.2	1.2 ± 0.3

Subclinical ketosis = 1.2 ≥ β-hydroxybutyrate ≤ 2.9 mmol/L

Clinical ketosis = β-hydroxybutyrate ≥ 3.0 mmol/L

*Milk composition at 10.2 ± 2.1 days postpartum

[a,b] Means not sharing the same superscript are significantly different (*P < 0.05*)

seems to alter other GH-dependent hepatic processes such as regulation of gluconeogenesis (Bernabucci et al. 2010).

Thus, despite the diminished feed intake (Garner et al. 2017), plasma NEFA in dairy cows suffering heat-stress do not increase (Shwartz et al. 2009) due to an increment in basal insulin (Wheelock et al. 2010), as this hormone is a potent antilipolytic hormone (Vernon 1992). This may explain why heat-stressed animals have a reduced mobilization of adipose tissue triglycerides (Baumgard and Rhoads 2013) which would reduce the occurrence of spontaneous ketosis.

Low blood levels of BHB had a positive effect on the establishment of pregnancy at first service, which is contrary to findings of several authors who have found that SCK is not related to conception rate at first insemination (McArt et al. 2012; Shin et al. 2015). On the other hand, Rutherford et al. (2016) observed that first insemination was 4.3 times less likely to be successful in SCK cows compared with nonketotic cows. However, overall conception rates in the present study were similar between ketotic and nonketotic cows, which is not consistent with previous reports in which hyperketonemia was detrimental to overall conception rate (Walsh et al. 2007; Ospina et al. 2010c). The present results suggest that reproductive integrity was not affected by the rise of BHB early in lactation. This lack of difference in overall conception rate compared with previous studies seems to stem from the fact that non-pregnant cows were repeatedly re-bred (cows were served about five times per pregnancy) and is likely that by the time cows received their third service, the ketotic state had subsided.

Detection of SCK or CK at 6 to 15 days postpartum, with no treatment of positive cows, did not decrease 305-day milk yield. The association between SCK and milk yield has been contradictory, with associations both negative and positive (Rajala-Schultz et al. 1999; Ospina et al. 2010a, b). Such an

Table 3 Reproductive variables for Holstein cows diagnosed with subclinical or clinical ketosis in a large commercial dairy operation in a hot arid environment

Reproductive variables	No ketosis (control)	Subclinical ketosis	Clinical ketosis
Conception rate (all services), %	80 (196/244)	82 (53/65)	81 (28/35)
Conception rate at first service, %	19 (39/201)^{a}	8 (4/43)^{b}	4 (1/28)^{b}
Services per conception*	4.6 ± 3.4	5.3 ± 3.5	4.9 ± 3.6
Calving to 1st estrus interval, days	36 ± 14	34 ± 14	39 ± 9
Calving to 1st service interval, days	59 ± 23^{a}	67 ± 34^{b}	65 ± 20^{b}
Calving to conception interval, days	192 ± 79^{a}	217 ± 83^{b}	200 ± 80^{ab}
Fetal losses, %	11 (22/205)^{a}	19 (10/54)^{b}	11 (3/28)^{ab}
Involuntary cow culling, %	18 (43/244)	18 (12/66)	20 (7/35)

Subclinical ketosis = β-hydroxybutyrate > 1.2 and ≤ 2.9 mmol/L

Clinical ketosis = β-hydroxybutyrate ≥ 3.0 mmol/L

*Cows with a confirmed pregnancy diagnosis

[a,b] Means not sharing the same superscript are significantly different (*P < 0.05*)

ambiguous association stems from different times postpartum periods for SCK diagnosis, different parities of cows, different cutpoints for defining ketosis, blood or milk evaluation for ketone bodies, and, more important, the way of calculating milk production.

The association between high blood BHB levels and short-term milk yield has been well established (Duffield et al. 2009; McArt et al. 2012). Both Dohoo and Martin (1984) and Duffield et al. (2009) observed milk yield loss of 1 to 2 kg/day in cows with hyperketonemia. McArt et al. (2012) observed that for each 0.1 mmol/L increase in blood BHB above 1.2 mmol/L early in lactation, milk yield decreased by 0.5 kg/day in the first 30 days of lactation. However, when Duffield et al. (2009) projected milk yield of cows with serum BHB ≥ 1.4 mmol/L in week 2 postcalving, milk production of the ketotic cows was over 240 kg greater than healthy cows. In the present study, although not statistically significant, cows with CK produced on average 395 more kilograms of milk than healthy cows. This apparent contradiction seems to be explained by the fact that high milk-yielding cows are at greater risk of suffering ketosis because higher serum ketone bodies occur in early lactation as an adaptive response to negative energy balance. In fact, some studies have recognized previous lactation milk yield as a risk factor for CK (Fleischer et al. 2001). Also, it could be that hyperketonemia in cows in this study resulted in earlier resolution, due to suboptimum milk yield (West 2003) and milk solids (Van Laer et al. 2015), due to the heat stress experienced by cows.

Ketotic cows presented a higher peak milk yield than healthy cows. This seemingly contradictory result could be due to the fact that cows with higher initial milk yield show higher peak yield (Elahi Torshizi 2016) and consequently mobilized more body fat which led to ketosis. Also, it could be that by the time peak yield was reached, a complete remission of ketosis displayed at 6 to 15 days had occurred in high-yielding cows.

Conclusions

Higher temperature-humidity index at calving and parturitions during the warmest seasons (summer and autumn) were major protective factors for developing CK in Holstein cows in a hot environment. In addition, these data indicated that detection of SCK or CK at 6 to 15 days postpartum, with no treatment of positive cows, did not decrease overall conception rate or 305-day milk yield, although elevated circulating BHB were negatively associated with conception rate at first service and fetal losses. Thus, it appears that physiological and metabolic adjustments for acclimation to heat stress in Holstein cows limit the occurrence of hyperketonemia and decrease the impact of this condition on 305-day milk yield and overall reproductive performance.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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